



JIMMA INSTITUTE OF TECHNOLOGY
FACULTY OF COMPUTING AND INFORMATICS
COMPUTER SCIENCE PROGRAM
FINAL YEAR PROJECT-1

“SEMI” | “አሙ” VOICE TO SIGN LANGUAGE TRANSLATOR

Group Member

ID No.	Name	Advisor's name
RU2976/13	MEKLIT DARGE	Mr. Samuel S. _____
RU3395/13	SAMUEL TESHOME	
RU2532/13	SINGSONG ADDISU	Mr. Mikru L. _____
RU3129/13	SINTAYEHU DESALEGN	

A Final Year project or proposal

Submitted to Faculty of Computing and Informatics, Jimma Institute of Technology, JimmaUniversity, in Partial fulfillment of the requirement of the Degree of Bachelor of Science in **Computer science**.

Jimma. Ethiopia

January 2024

Acknowledgment

We would like to express our profound gratitude to the Almighty for His continuous blessings and guidance throughout the journey of our final year project. Our deepest appreciation goes to our esteemed advisors, Mr. Mikru L. and Mr. Samuel S., whose unwavering support, insightful guidance, and expertise have played a pivotal role in shaping and ensuring the success of our project. Our sincere thanks extend to Jimma University for providing us with the opportunity to undertake this project. The university's commitment to fostering innovation and technological advancements has been a driving force in our academic journey.

Heartfelt appreciation is also extended to our fellow classmates and colleagues whose valuable input and contributions have significantly enriched our project.

This acknowledgment is a reflection of the collaborative efforts and support that have made our final year project a reality. Thank you to everyone who has been a part of this fulfilling experience.

Contents

Acknowledgment	i
Abstract	vi
Abbreviations	vii
Chapter One Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Statement of the problem	3
1.4 Objective of the project	4
1.4.1 General Objective	4
1.4.2 Specific Objectives	4
1.5 Feasibility	4
1.5.1 Economic feasibility	5
1.5.2 Technological Feasibility	5
1.5.3 Time Feasibility	5
1.5.4 Behavioral/Political Feasibility	6
1.5.5 Market Feasibility	6
1.5.6 Schedule Feasibility	6
1.6 Scope of the project	6
1.6.1 Scope of the Project	6
1.7. Significance of the Project	6
1.7.1. Inclusive Education	7
1.7.2. Professional Empowerment	7
1.7.3. Technological Innovation for Social Good	7
1.7.4. Bridging Communication Gaps	7
1.7.5. National and Global Relevance	8
1.8 Target Beneficiaries	8
1.9 Methodology	9
1.9.1 Data Source and Preprocessing	9
1.9.2 Fact-Finding Techniques	10
1.9.3 Systems Analysis and Design	10
1.9.4 Development Tools	10
1.9.5 Training and Evaluation	12

1.9.6 Implementation (Direct):.....	12
1.10 Limitations of the Project.....	12
Chapter Two Requirement Analysis and Specification	13
2.1 Introduction of Existing/Current System:	13
2.1.1 Challenges in the Current System	14
2.2 Players in the Existing System	14
2.3 Business Rules	15
2.4 Proposed System	15
2.5 Requirements of the Proposed System	16
2.5.1 Functional Requirements:	16
2.5.2 Nonfunctional Requirements:	16
2.6 Analysis Model	18
2.6.1. Use case model.....	18
2.6.2 Use case documentation.....	20
2.6.3 Sequence diagram	21
2.6.4 State chart diagram	22
2.6.5 Activity diagram.....	23
2.6.6 Class Diagram	25
Chapter -Three: System Design.....	26
3.1. INTRODUCTION.....	26
3.2. Purpose of the System.....	26
3.3. Design Goals	26
3.4. Deployment Diagram.....	27
3.5. Architectural Design	28
3.5.1. Subsystem decomposition.....	28
3.5.2. Component diagram	29
3.5.3. Persistent modeling	30
3.5.4. Access Control and Security.....	31
3.5.5. Global Software Control.....	32
3.5.6. Boundary Conditions.....	33
3.5.7. Machine learning model design	34
3.6. User Interface Design.....	35
3.7. Database Design	38

3.8. Machine learning algorithm.....	38
3.9. Summary.....	39
References.....	39

Table of Figueres

FIGURE 1 USE CASE DIAGRAM	19
FIGURE 2 SEQUENCE DIAGRAM	21
FIGURE 3 STATE DIAGRAM FOR LOGIN	22
FIGURE 4 STATE DIAGRAM FOR USER	22
FIGURE 5 STATE CHART	23
FIGURE 6 ACTIVITY DIAGRAM	24
FIGURE 7 CLASS DIAGRAM	25
FIGURE 8 DEPLOYMENT DIAGRAM.....	28
FIGURE 9 COMPONENT DIAGRAM	30
FIGURE 10 MACHINE LEARNING MODEL DESIGN	34
FIGURE 11 HOME PAGE DEFAULT DISPLAY	36
FIGURE 12 HOME PAGE DISPLAY WHEN RECORDING.....	36
FIGURE 13 FEEDBACK PAGE.....	37
FIGURE 14 LOGIN PAGE	37

List of Table

Table 1 use case documentation	20
--------------------------------------	----

Abstract

"ሰሚ" | "SEMI" is a groundbreaking project that seeks to revolutionize inclusivity in educational and professional settings for individuals with hearing impairments. The motivation behind this endeavor arises from the persistent communication challenges faced by hearing-impaired students in our class. This project introduces a transformative solution an audio to sign language translator.

Our technology takes audio input, utilizes Google APIs for speech recognition, displays the recognized text, and generates sign language representations. The project envisions a future where every voice is heard and understood, breaking down communication barriers and promoting inclusivity through innovative technology. "ሰሚ" | "SEMI" sets a new standard for education and professional communication, fostering diversity and enriching the learning experience.

Abbreviations

- Hard of hearing – HoH
- Ethiopian Sign language – EthSL
- Machine Translation – MT
- Natural language processing – NLP
- Convolutional Neural Networks - CNN
- Role-Based Access Control – RBAC
- Hypertext transfer protocol – HTTPS
- object-relational mapping - ORM
- Secure Socket Layer - SSL
- Role-Based Access Control – RBAC
- Application Programming Interface - API
- Transport Layer Security - TLS

Chapter One Introduction

1.1 Background

Communication is a way of sharing information between people to understand each other better. Deaf and Hard-of-Hearing (HoH) individuals, who cannot hear well or at all, have the right to access information like everyone else. However, they face challenges due to limited ways of communicating, leading to difficulties in understanding and being understood by others. Over 5% of the world's population – or 430 million people – require rehabilitation to address their disabling hearing loss (432 million adults and 34 million children). It is estimated that by 2050 over 700 million people – or 1 in every 10 people – will have disabling hearing loss.

“ሰሚ” | “SEMI” project aims to help the Deaf and HoH community in Ethiopia overcome communication barriers. While many people use spoken language to communicate, Deaf and HoH individuals rely on non-verbal communication, especially sign language, to express themselves.

Sign language is a visual language where information is conveyed through hand shapes, motions, facial expressions, and other visual cues. In Ethiopia, Deaf and HoH people use Ethiopian Sign Language (EthSL), which has unique features that make communication challenging. Unlike spoken languages, sign language lacks universal recognition systems, making it harder for Deaf and HoH individuals to communicate and access information.

Machine Translation (MT) systems, which are computer programs translating voice or speech from one language to another, have become preferable over human interpreters for various reasons. These include the need for large-scale translation, dealing with technical materials, ensuring consistency, increasing translation speed, and reducing costs.

Despite extensive research on voice to sign language conversation, there are limited projects focusing on EthSL. The existing solutions are not well-developed or widely introduced to help students and professionals with hearing impairment. Therefore, the main goal of this project is to create and introduce a system that can translate spoken language into Ethiopian Sign Language, making communication more accessible for the Deaf and HoH community in Ethiopia.

1.2 Motivation

In the realm of communication, sign language serves as an intricate and fully developed means of interaction among Deaf and hearing signers. However, a noticeable communication gap persists between Deaf individuals and those without signing skills. In attempting to bridge this gap, hearing individuals often resort to text-based communication, which, unfortunately, proves unsatisfactory, especially when engaging with pre-lingual Deaf individuals—those who predominantly possess poor to moderate reading skills.

The challenge is further compounded by the scarcity of hearing individuals proficient in sign language, necessitating the presence of interpreters in various activities involving Deaf individuals. This reliance on interpreters imposes limitations on the active participation of Deaf individuals in a multitude of settings.

Our motivation for the "ሰሜን/ SEMI" project is deeply rooted in the firsthand experiences of students within our academic community. The presence of hearing impairment creates a substantial barrier to communication and inclusion, particularly within the confines of our classrooms. The struggles observed among hearing-impaired students have not only fueled our commitment to find a solution but have also kindled a passion to address a broader, national issue in Ethiopia.

Our motivation extends beyond being a mere reaction to a local problem; it represents a call to action on a national scale. By addressing the communication challenges faced by the hearing-impaired community, we aspire to contribute to a transformative solution that ensures equal learning opportunities for all, breaking down barriers to inclusion and fostering a more inclusive educational landscape in Ethiopia.

1.3 Statement of the problem

In the educational institutions of Jimma Institute and others across Ethiopia, a critical challenge persists, significantly impacting the academic journey of hearing-impaired students. At the heart of this issue is a pronounced communication barrier, accentuated by the scarcity of vital resources, particularly proficient sign language interpreters. The existing communication methods within Ethiopia do not adequately cater to the unique needs of students with hearing impairments. Consequently, these students face isolation within the academic environment, unable to actively participate in classroom discussions, access educational materials, or engage meaningfully with their peers.

The deficiency in inclusive communication strategies hampers the educational experience for hearing-impaired students, impeding their right to equitable learning opportunities. The “ሰሜን” | “SEMI” project emerges as a response to this pressing problem, propelled by the belief that education should be accessible to every student, irrespective of their hearing abilities. The project seeks to dismantle the barriers that hinder the full engagement of hearing-impaired students in the learning process.

By acknowledging the existing limitations and challenges faced by hearing-impaired students, the project aspires to create an educational environment where these students can seamlessly participate in classroom activities, contribute to discussions, and access educational content without hindrance. The core objective is to transform the educational landscape by developing a real-time sign language conversion platform, addressing the communication disparities and fostering inclusivity for all students, regardless of their hearing abilities. The significance of this endeavor extends beyond individual educational institutions; it encompasses a broader vision for an inclusive education system in Ethiopia.

1.4 Objective of the project

1.4.1 General Objective

The general objective of our project is to develop a voice to Ethiopian sign language translator system for deaf and hard of hearing students and professionals.

1.4.2 Specific Objectives

- ✓ Develop a robust system to capture and process voice input data for effective voice recognition in various linguistic contexts.
- ✓ Train a sophisticated model capable of converting voice input into accurate and contextually appropriate sign language gestures in Amharic.
- ✓ Systematically organize Amharic and corresponding Ethiopian Sign Language (EthSL) text structures at both the word and letter levels to ensure coherence and precision in translation.
- ✓ Design and implement a prototype model that facilitates seamless communication between individuals with hearing impairments and those without, enhancing accessibility and fostering effective interaction.
- ✓ Test our system
- ✓ Explore and integrate the system with existing educational tools and platforms, enabling seamless incorporation into educational settings to support learning and communication for hearing-impaired students and educators.

1.5 Feasibility

Feasibility is designed to answer whether or not a proposed project or idea should go forward by determining whether the project or plan is practical. A feasibility assessment is a test of a system according to its workability, impact, or the organization's ability to meet user needs, and effective user resources. It focuses on three major questions:

- What are the user needs?
- What resources are available for the system to be designed?
- What are the impacts of the system on us to design?

1.5.1 Economic feasibility

Economic feasibility more determines the benefits and the savings that are expressed from a proposed system and compares those costs.

For developing our system, the only needs are: -

- Personal computer,
- camera, Wi-Fi and
- Person who knows the sign language well.

Therefore, since it doesn't need much cost for development, the system is economically feasible for development. Unfortunately, most of the hearing and speech impaired people doesn't use smart phone or other communication devices for communication purpose, to use this system they use the organization's device for communication purpose.

1.5.2 Technological Feasibility

The assessment of this feasibility must be based on an outline design of the system requirements in terms of input, output, programs, and procedures. Technical feasibility centers in the existing computer, to what extent it supports the proposed addition and accessing Wi-Fi is enough for achieve our project. Since we don't need another technology, our project is technologically feasible.

1.5.3 Time Feasibility

Every operational activity of our project is scheduled well and analyzed deeply to make every work of the system efficient and complete as per the defined schedule. In the beginning what should be done first like gathering information and the things to be performed next will be planned properly. The system is fast and accurate to convert Sign language to Text. Because we will use optimal amount of image for training our model, standard resolution pixel for label our input data. So, our project has time feasibility.

1.5.4 Behavioral/Political Feasibility

Behavioral and political feasibility explores the acceptance and support of the project within the user community and political environment. This aspect will be assessed based on the willingness of users to adopt the technology and any potential political factors influencing project implementation.

1.5.5 Market Feasibility

Market feasibility evaluates the system's potential success in the market. It will consider factors such as the demand for the technology, competition, and potential barriers to market entry.

1.5.6 Schedule Feasibility

Schedule feasibility examines the realistic timeline for project completion. It ensures that each phase is appropriately planned and executed to meet the defined schedule.

1.6 Scope of the project

1.6.1 Scope of the Project

This project involves creating a Progressive Web App (PWA) designed to enhance communication by converting spoken words into Ethiopian Sign Language. The system employs a speech-recognition system to recognize speech, converting it into proper sentence structures, words, and alphabets. The development process includes selecting an appropriate neural network architecture, creating datasets (development, training, and test datasets), and training the model using the training dataset. Through fine-tuning with different values on the development dataset, we aim to identify optimal parameters and hyperparameters. The final step involves evaluating the model's performance using relevant metrics on the test dataset.

1.7. Significance of the Project

The “ሰሚ” | “SEMI” project holds immense significance, addressing a critical need and contributing to transformative change in the lives of hearing-impaired individuals. The unique challenges faced by this community underscore the importance of the project in several key aspects.

1.7.1. Inclusive Education

In Ethiopia, hearing-impaired students encounter formidable communication barriers that impede their full participation in educational activities. The “ስሚ” | “SEMI” system serves as a catalyst for inclusive education, breaking down these barriers and ensuring that every student, regardless of their hearing abilities, can actively engage in classroom discussions, lectures, and other academic interactions. By fostering an inclusive educational environment, the project paves the way for a more equitable and enriching learning experience for all.

1.7.2. Professional Empowerment

Beyond the academic sphere, the project extends its impact to the professional realm. Hearing-impaired individuals often face challenges in workplace communication, hindering their professional growth and opportunities. “ስሚ” | “SEMI” empowers these individuals by facilitating effective communication in various professional settings, including workplaces, conferences, and public events. This not only enhances their career prospects but also contributes to the diversity and inclusivity of Ethiopia's professional landscape.

1.7.3. Technological Innovation for Social Good

The project exemplifies the transformative power of technology harnessed for social good. By leveraging advanced algorithms and user-friendly interfaces, “ስሚ” | “SEMI” demonstrates the potential of innovation to address real-world challenges and improve the lives of hearing-impaired individuals in Ethiopia. This technological innovation not only meets a specific need but also sets a precedent for the use of technology in creating positive social impact.

1.7.4. Bridging Communication Gaps

One of the most significant contributions of the project is its ability to bridge communication gaps between the hearing-impaired community and the broader society. With “ስሚ” | “SEMI”, communication becomes more accessible, efficient, and meaningful. This has broader implications for fostering understanding, empathy, and collaboration between individuals with varying hearing abilities, creating a more connected and inclusive social fabric in Ethiopia.

1.7.5. National and Global Relevance

The project's significance extends beyond national borders, positioning Ethiopia as a pioneer in inclusive technology solutions. As the global community increasingly recognizes the importance of inclusivity, "ሰሚ" | "SEMI" " serves as a model for other regions facing similar challenges. Its success has the potential to influence global perspectives on technology-driven solutions for hearing-impaired individuals, showcasing Ethiopia's commitment to innovation and inclusivity on the international stage.

1.8 Target Beneficiaries

The "ሰሚ" | "SEMI" " system is designed with a commitment to inclusivity, aiming to benefit a diverse range of users within educational and professional settings. The primary target beneficiaries include:

1. Hearing-Impaired Students:

- Immediate access to educational content.
- Active participation in classroom activities, discussions, and lectures.
- Enhanced learning experiences through real-time communication.

2. Educators:

- Improved communication with hearing-impaired students.
- User-friendly interface for seamless integration into educational environments.
- Tools for fostering inclusivity and diverse learning experiences.

3. Professionals in Various Fields:

- Accessibility to workplace communication.
- Participation in conferences and public events with ease.
- Enhanced opportunities for collaboration and engagement.

4. General Public:

- Increased awareness and understanding of sign language.
- Contribution to a more inclusive and diverse society.

5. Organizations and Institutions:

- Facilitation of inclusive educational environments.
- Demonstration of commitment to diversity and accessibility.

6. Future Developers and Innovators:

- Source of inspiration for developing similar technologies.
- Potential for further innovation in the field of assistive technologies.

The "ሰሜን" | "SEMI" " system aims to empower individuals with hearing impairments, creating a positive impact on their educational, professional, and social interactions. The ripple effect of this technology extends beyond its immediate users, fostering a more inclusive and understanding society at large.

1.9 Methodology

The methodology employed for developing the "ሰሜን" | "SEMI" " project, centered around voice-to-sign language conversion, is grounded in a series of strategic steps and incorporates essential concepts to ensure the system's accuracy and effectiveness in real-world applications.

1.9.1 Data Source and Preprocessing

- **Data Type**

Define the type of data required for the voice-to-sign language converter project, emphasizing spoken Amharic phrases and sentences.

- **Data Source**

Source primary data from a diverse group of native Amharic speakers, capturing varied accents and speech patterns.

- **Data Cleaning**

Implement data cleaning processes to address noise, inconsistencies, and outliers in the collected dataset.

- **Data Labeling or Annotation**

Label and annotate the dataset, specifying each sign language gesture corresponding to its spoken counterpart, creating a labeled dataset for training.

1.9.2 Fact-Finding Techniques

- **Interview**

Engaging with experts in sign language through interviews provides valuable insights into the intricacies of sign language structure and grammar. These insights guide the system's training to align with real-world signing variations.

- **Practical Observation**

Real-world observations of sign language gestures, facilitated through in-person interactions and recorded videos, contribute to the system's comprehension of nuanced signing expressions.

- **Document Analysis**

Leveraging existing documentation, including sign language dictionaries and linguistic resources, enriches the system's understanding of sign language vocabulary and linguistic conventions.

1.9.3 Systems Analysis and Design

The architecture of the system is meticulously designed to process diverse audio inputs, employing algorithms for translation while prioritizing user-friendliness and real-time responsiveness. Concepts from natural language processing and computer vision are integrated to enhance system capabilities.

1.9.4 Development Tools

For the development phase of the voice-to-sign language converter, careful consideration has been given to selecting tools that facilitate efficient model creation, training, and integration. The chosen tools contribute to the overall effectiveness of the project:

- **Programming Language: Python (3.10)**

Why Python: Python is selected as the primary programming language due to its open-source nature, ease of development, and extensive support for machine learning frameworks. Its accessibility to various AI libraries and frameworks makes it a versatile choice for implementing the voice-to-sign language converter.

- **Machine Learning Frameworks: TensorFlow, Keras**

Why TensorFlow: TensorFlow is employed as the core machine learning framework due to its robust capabilities in building and training neural networks. Keras, being a high-level neural networks API running on top of TensorFlow, complements the development process by providing a user-friendly interface for model design and deployment.

- **Computer Vision Library: OpenCV**

Why OpenCV: OpenCV is integrated for computer vision tasks, such as image preprocessing and feature extraction. Its broad set of tools and algorithms simplifies image manipulation, making it a suitable choice for handling the dataset and enhancing model input.

- **Git GitHub**

Why Git and GitHub: Version control systems like Git and platforms like GitHub can greatly facilitate collaborative development and project management. They allow you to track changes, collaborate with others, and easily manage different versions of your codebase

- **Other Applications: Webcams, MS-PowerPoint, Notepad, Chrome, YouTube Browsers**

How They Are Used: Webcams are utilized for real-time data collection, capturing additional datasets for diverse scenarios. MS-PowerPoint aids in creating documentation and presentations. Notepad serves as a lightweight text editor for code and data manipulation. Chrome and YouTube browsers are used for research, reference, and editing files.

This comprehensive set of development tools is meticulously chosen to ensure a streamlined and effective development process for the voice-to-sign language converter, encompassing model building, training, evaluation, and collaborative project management.

1.9.5 Training and Evaluation

we're shaping our voice-to-sign language converter to be both sharp and reliable. Here's how:

Training/Testing Data Split:

We're teaching our model like a student preparing for an exam. The majority of our data forms the "study material" (training dataset), while a smaller chunk acts like the unseen questions in the exam (testing dataset). Striking a good balance here helps our model handle different situations with ease.

Performance/Evaluation:

After the "study," our model faces a performance review. Think of it as grades for accuracy and precision. This review helps us fine-tune our model, making sure it's ready for real-life challenges like a pro sign language translator.

1.9.6 Implementation (Direct):

Now, for the big move! We're opting for a direct implementation strategy, which means we're replacing any existing system with ours, no middle steps. We're confident that our model is up to the task, and we want users to experience the upgrade right away. It's like switching to a new and improved system without a long adjustment period. This approach requires careful testing to ensure a smooth and trouble-free transition.

1.10 Limitations of the Project

While the "ሰሚ" | "SEMI" project aims to address critical communication gaps, it is important to acknowledge certain limitations that may impact its scope and implementation. These limitations include language scope, hardware dependency, training data variability, environmental noise, internet connectivity, adaptation to local dialects, user learning curve, EthSL lexical expansion, limited accessibility to technology, and security and privacy concerns. By recognizing these limitations, the project team can proactively work towards mitigating them to ensure the system's effectiveness and maximize its impact.

Chapter Two Requirement Analysis and Specification

2.1 Introduction of Existing/Current System:

In Ethiopia, the hearing-impaired community faces formidable communication barriers due to a lack of accessible channels. Unfortunately, the deaf community has been largely overlooked, despite their significant population.

Currently, Ethiopian Sign Language serves as the primary means of communication for the deaf community in Ethiopia. However, the general population's limited understanding of sign language poses challenges for the deaf society, making it difficult for them to express themselves in various contexts and hindering their full participation and inclusion.

In terms of existing systems, there is a lack of effective solutions in Ethiopia that convert spoken language into sign language in real-time. While some research has been conducted at the national level, practical implementation of such a system is yet to be realized.

Furthermore, another significant challenge faced by the hearing-impaired community is the absence of sign language interpreters in Ethiopian universities, except for Addis Ababa University. This lack of interpreters presents a major barrier for hearing impaired students who aspire to attend universities outside of Addis Ababa. The limited availability of sign language interpretation services hinders their access to higher education, making it challenging for them to fully participate in academic settings and hindering their academic and professional growth.

Currently, the existing system relies predominantly on manual interpretation by sign language interpreters, which is limited in availability and can lead to communication delays and inaccuracies. The absence of a reliable and efficient voice-to-sign language conversion system further hinders effective communication and access to information for the hearing-impaired community.

2.1.1 Challenges in the Current System

Geographical Disparities: Even in instances where personal interpreters are accessible, their availability is not uniform across regions and institutions. This geographical imbalance exacerbates the communication gap, hindering Deaf individuals in certain areas from enjoying equal participation in various aspects of life.

Inadequacies in Organizational Support: The current reliance on personal interpreters lacks a structured organizational framework, particularly outside a select few universities and organizations. This deficiency limits the comprehensive support needed for effective communication in diverse settings.

Absence of sign language interpreters in universities: The lack of sign language interpreters in Ethiopian universities, except for Addis Ababa University, creates a significant barrier for hearing impaired students who wish to pursue higher education outside of the capital city. This limitation hampers their ability to fully participate in academic settings and inhibits their academic and professional development.

2.2 Players in the Existing System

In the current system, the primary players involved in facilitating communication for individuals with hearing impairments include:

Personal Interpreters: Personal interpreters play a crucial role in facilitating communication between hearing-impaired individuals and the hearing world. However, their availability is limited, and their deployment is often concentrated in specific academic institutions or organizations. This scarcity poses challenges for widespread accessibility.

Educational Institutions: Some universities and educational institutions in Ethiopia may have arrangements for personal interpreters to assist hearing-impaired students. However, the coverage is uneven, leaving many students without adequate support, especially in remote or underprivileged areas.

Organizations Supporting the Hearing-Impaired: Certain organizations within Ethiopia focus on supporting the hearing-impaired community. While commendable, the resources and services offered by these organizations may not be universally accessible, contributing to a persistent communication gap.

2.3 Business Rules

Business rule of “ስሚ” | “SEMI” are

- Users must be registered with the system to access voice to sign language conversion services.
- Authorization from the administrator is a prerequisite.
- The system operates under the principle that accurate and accessible sign language translations should be provided for effective communication between users.
- Security protocols are paramount to protect the privacy of user data within the Voice to Sign Language Interpreter system.
- The user interface of the platform, encompassing both web and mobile applications, is designed to comply with recognized accessibility guidelines and standards.
- The system incorporates advanced algorithms to ensure precise and timely translation of spoken language into sign language.
- Users, particularly Deaf Students, can expect immediate access to the sign language interpretation of voice input.
- The system administrators have the authority to manage and configure the platform to cater to the evolving needs of the educational and technological landscape, ensuring sustained relevance and impact.

2.4 Proposed System

Proposed System Description:

The envisioned “ስሚ” | “SEMI” is a revolutionary system designed to surmount the challenges prevalent in the existing communication landscape for the hearing-impaired. Unlike the current limited reliance on personal interpreters in select institutions, “ስሚ” | “SEMI” introduces a voice-to-sign language conversion mechanism, ensuring real-time and accurate translation for seamless communication. This innovative approach not only eliminates the scarcity of interpreters but also extends inclusivity beyond academic settings, embracing a broader societal context. The system's user-friendly interface, advanced algorithms, and educational inclusivity features collectively form an alternative that transcends the limitations of the current system, fostering a more connected and inclusive environment for the hearing-impaired community.

2.5 Requirements of the Proposed System

2.5.1 Functional Requirements:

The “ሰሚ” | “SEMI” system is meticulously designed to meet a spectrum of functional requirements, ensuring a seamless and enriching experience for its users.

❑ Performance requirements:

Real-time Conversion: The system should be able to convert spoken language into sign language in real-time to ensure immediate and seamless communication.

❑ Process requirements:

Speech Recognition: The system should have efficient and accurate speech recognition capabilities to convert the spoken language into text.

❑ Input-related requirements:

Voice Input: The system should be able to capture voice input from the user.

❑ Output-related requirements:

Sign Language Generation: The system should generate sign language animations or avatars based on the converted text.

Text Display: The system should display the converted text in a clear and easily readable format.

These functional requirements emphasize the need for efficient processing, accurate speech recognition, capturing voice input, generating sign language animations, displaying text, providing audio output, and managing storage for user preferences and existing models. By fulfilling these requirements, the system can ensure optimal performance, seamless communication, and a user-friendly experience for individuals with hearing impairments.

2.5.2 Nonfunctional Requirements:

In crafting the architecture of the “ሰሚ” | “SEMI” system, attention to nonfunctional requirements is paramount, ensuring a holistic and user-centric design.

Performance:

The system must exhibit optimal performance even under varying loads and usage patterns. Response times for voice-to-sign language translation should be consistently swift, providing a seamless communication experience.

User Interface:

A user-friendly interface is imperative for a positive user experience. The system should feature an intuitive and accessible design, accommodating users with varying levels of technological proficiency. Clear navigation and comprehensible visual elements contribute to the system's overall usability.

Security and Access Permissions:

Robust security measures are integral to safeguard user data and system integrity. Access to sensitive information, both voice data and translated content, should be restricted based on secure authentication protocols. Encryption and secure data transmission must be prioritized to uphold user privacy.

Backup and Recovery:

The system should implement robust backup and recovery mechanisms to prevent data loss in unforeseen circumstances. Regular backups of user preferences, learning models, and system configurations are essential for maintaining continuity and minimizing disruptions.

Scalability:

Anticipating future growth, the system should demonstrate scalability. As user demands increase, the infrastructure and algorithms should seamlessly scale to accommodate higher loads without compromising performance.

Reliability:

Reliability is fundamental to user trust. The system should operate consistently without unexpected downtimes or errors. Redundancy and failover mechanisms should be in place to ensure uninterrupted service.

By meticulously addressing requirements, the "ሰሚ" | "SEMI" system aspires to redefine communication accessibility for the hearing-impaired, promoting a more inclusive and interconnected society with a secure, user-friendly, and reliable communication platform for users across diverse contexts.

2.6 Analysis Model

Analysis models are graphical or textual representations that help to understand, specify, design, and document a system. These models help in analyzing the system requirements and in communicating the system design to stakeholders. They are used during the analysis phase of the software development life cycle. There are several types of analysis models that can be used, including Activity Diagrams, Sequence Diagrams, Use Case Diagrams, and Class Diagrams. The choice of which models to use depends on the system requirements and the development methodology being used. Each model has its own strengths and can be used to represent different aspects of the system.

2.6.1. Use case model

Use case Diagram is a generalized description of how a system will be used. It provides an overview of the intended functionality of the system. It composes of the following major components: -

- System boundary
- Actors
- Use cases

System boundary: a rectangle diagram representing the boundary between the actors and the system.

Actors: A role that a user plays with respect to the system, including human users and other systems. The basic actors that take part in our system are the user and the system.

Use case: A set of scenarios that describing an interaction between a user and a system.

A use case is a single unit of meaningful work.

Each Use Case has a description which describes the functionality that will be built in the proposed system.

- **Include:** a dotted line labeled <<include>> beginning at base use case and ending with an arrow pointing to the include use case. An “Include” relationship is used to indicate that a particular Use Case must include another use case to perform its function.

- **Extend:** a dotted line labeled <<extend>> with an arrow toward the base case. The extending use case may add behavior to the base use case. The base class declares “extension points”.

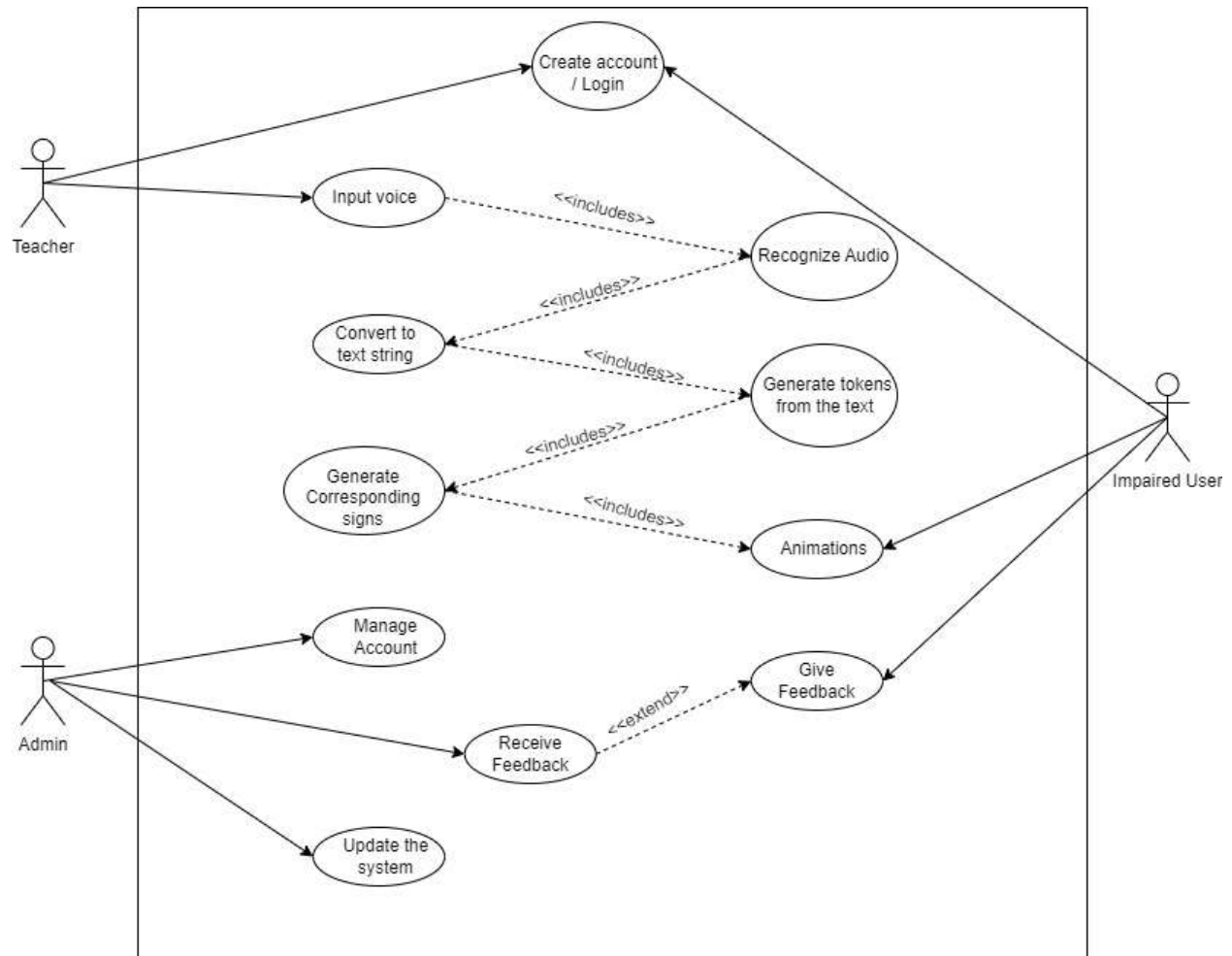


Figure 1 *use case diagram*

2.6.2 Use case documentation

Case ID.	UC:1
Name	Input voice
Actor	Teacher
Description	Impaired User Will Adjust the Mic to get audio input.
Pre-Condition	The Mic is already loaded
Post-Condition	Voice input will be recognized
Case ID.	UC:2
Name	Convert voice input to text
Actor	Speech recognition
Description	Recognizes the audio input and converts them to text with sentence, word and character structure.
Pre-Condition	The must be audio input.
Post Condition	Text format of recorded voice
Case ID.	UC:3
Name	Generate corresponding signs
Actor	Sign language generator
Description	By recognizing the converted voice to text, it will start doing mapping to the equivalent sign language representation.
Pre-Condition	Text recognition
Post-Condition	Sign language representation
Case ID	UC:4
Name	animation
Actor	Impaired User
Description	After The sign language generator generates sign language there will be a sign language display for the user.
Pre-Condition	Generated sign language
Post-condition	Animation display
Case ID	UC:5
Name	Manage accounts
Actor	Admin
Description	The admin will manage accounts which opened by the user of our system.
Pre-condition	Accounts must be opened
Post-condition	Users account will be managed and protected.

Table 1 use case documentation

2.6.3 Sequence diagram

A sequence diagram is a type of interaction diagram that shows the interactions between objects or components in a system in a particular sequence or order. It shows the messages that are exchanged between the objects or components and the order in which they occur.

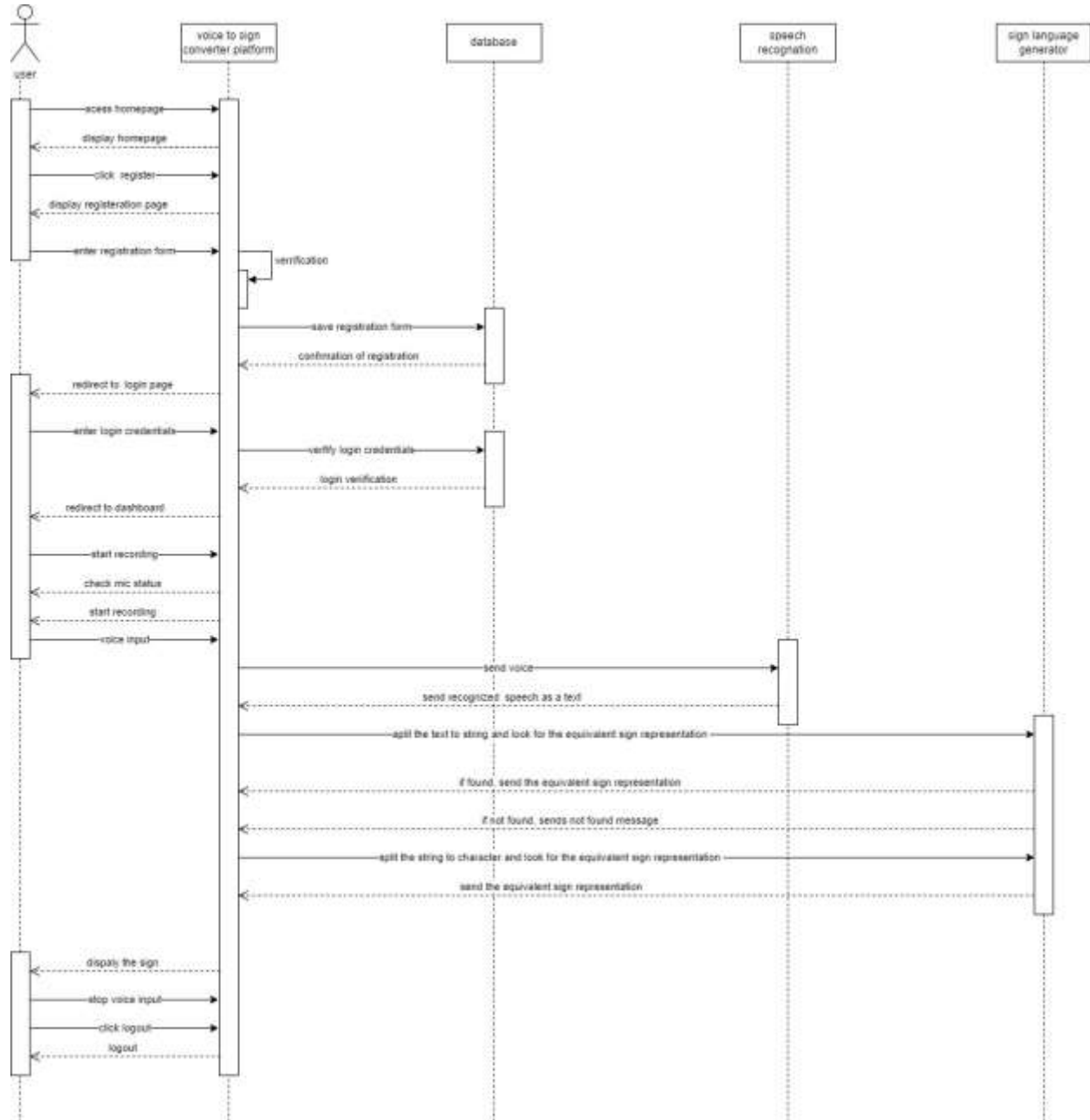


Figure 2 Sequence diagram

2.6.4 State chart diagram

A state chart diagram is a behavioral model that represents the different states that an object or system can be in, and the events or conditions that cause it to transition from one state to another. It shows the behavior of an object or system over time, and is useful for modeling systems with complex state behavior.

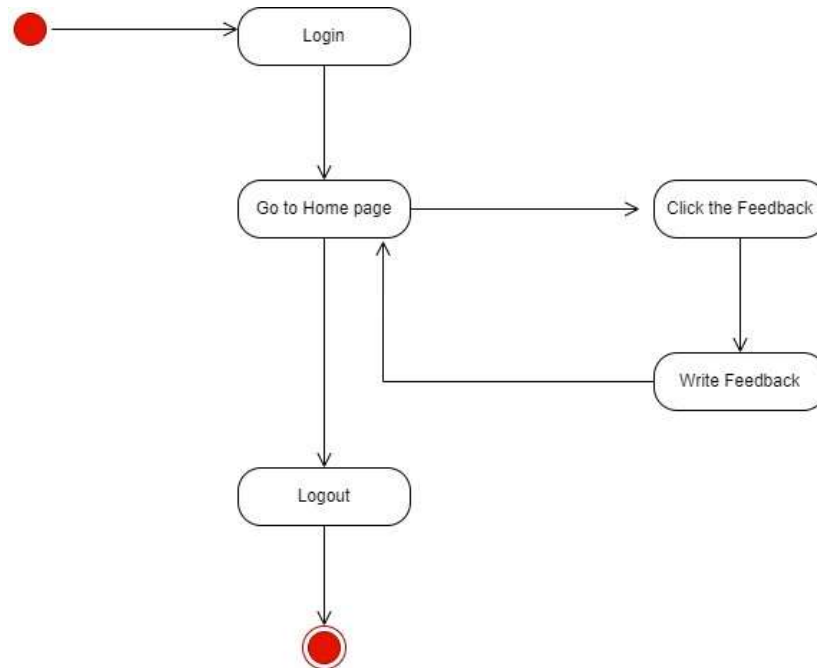


Figure 3 state diagram for login

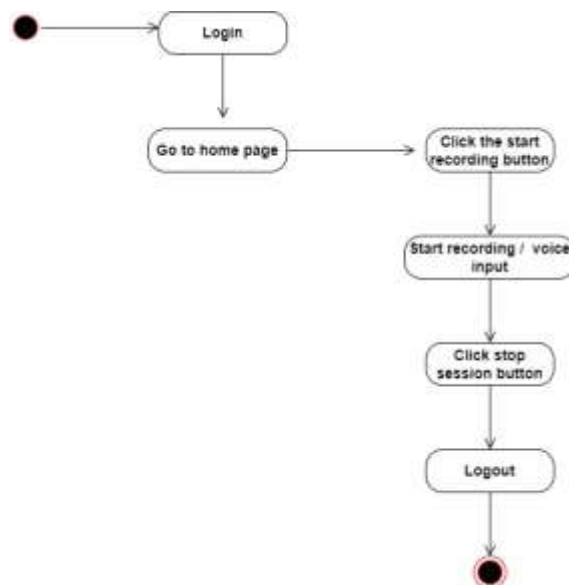


Figure 4 state diagram for user

The whole system state diagram

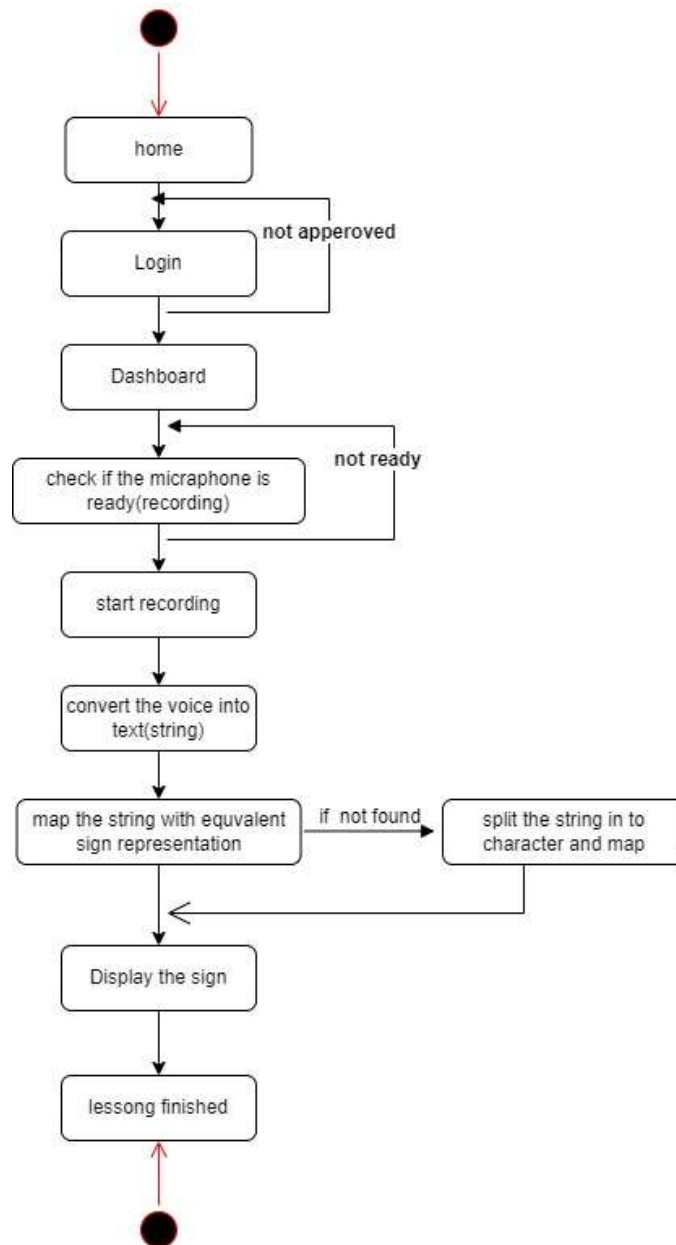


Figure 5 state chart

2.6.5 Activity diagram

An activity diagram is a type of behavioral diagram that represents the flow of activities or tasks in a system or process. It shows the actions or tasks that need to be performed, the decisions that need to be made, and the flow of control between different parts of the system or process.

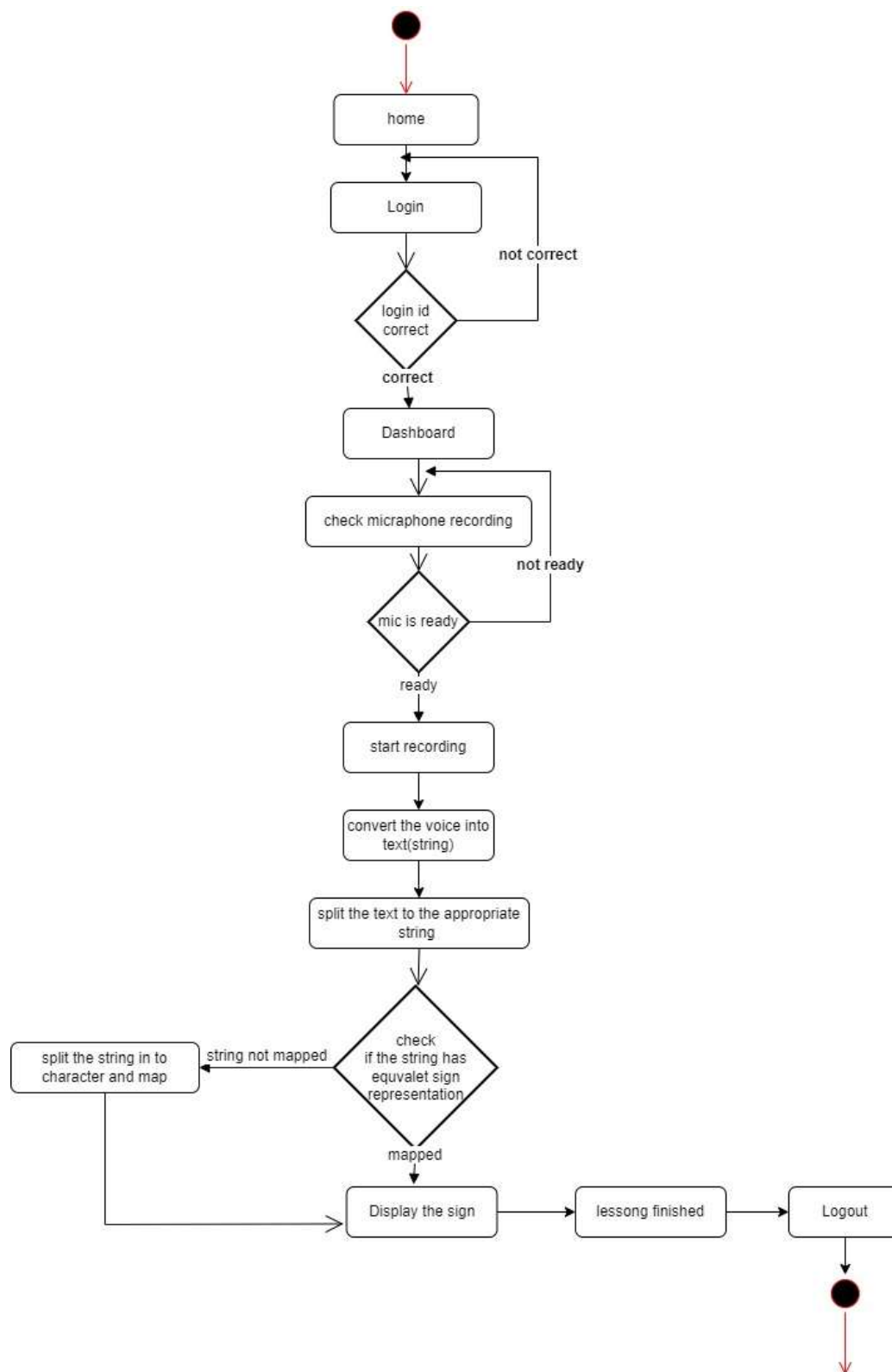


Figure 6 activity diagram

2.6.6 Class Diagram

A class diagram serves as a static model elucidating the structural composition of the “ሰሚ” | “SEMI” system, focusing on classes, interfaces, and their interconnections. It delineates the attributes and methods of each class, emphasizing:

Sign language generation system: Represents the system that generates the equivalent sign language which is equivalent to the voice input.

Speech Recognition System: Utilizes advanced algorithms to convert voice input into text.

Text To Sign Language Converter: Transforms textual information into sign language animations or videos.

Deaf Student: Primary user class benefiting from the conversion of speech to sign language.

Educational institute: Educators interacting with the system to deliver lessons using speech or voice.

Administrator: Responsible for system management, ensuring smooth operation and handling technical issues.

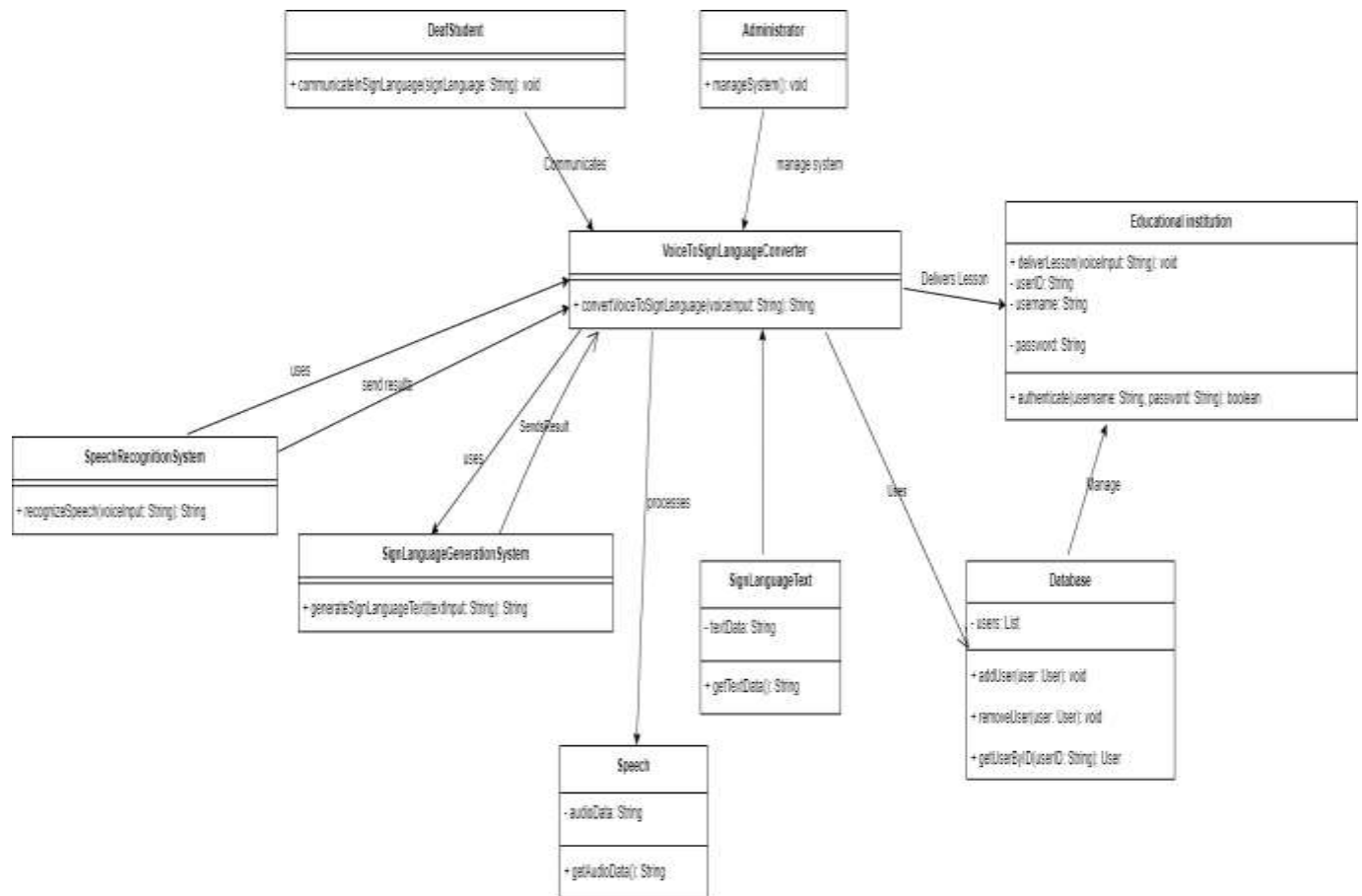


Figure 7 class diagram

Chapter -Three: System Design

3.1. INTRODUCTION

Within the realm of fostering inclusive education, the development of an innovative platform takes center stage. This system endeavors to bridge communication barriers by translating spoken language into text and further transforming it into dynamic sign language animations. This section delves into the fundamental objectives propelling the creation of this transformative system, specifically designed to aid deaf students in accessing classroom materials effectively.

3.2. Purpose of the System

The primary objective of this system is to empower deaf students within educational environments. By harnessing cutting-edge speech recognition and animation technologies, the platform aims to seamlessly convert spoken language into coherent textual content and subsequently into vibrant Ethiopian sign language animations. Its ultimate purpose is to facilitate comprehension and engagement with class materials through a preferred mode of communication for deaf students.

3.3. Design Goals

Comprehensive Design Goals:

- ✓ **Seamless Conversion:** Enable rapid and accurate transformation of spoken language into textual form, ensuring precision and clarity in the interpretation process.
- ✓ **Authentic Sign Language Representation:** Generate sign language animations that authentically reflect the converted text, emphasizing accuracy in hand gestures, expressions, and linguistic nuances.
- ✓ **User-Centric Interface:** Develop an intuitive and accessible user interface, allowing effortless interaction for users to input voice, view text, and engage with sign language animations seamlessly.
- ✓ **Real-Time Animation:** Achieve real-time generation and display of synchronized sign language animations alongside the converted text, promoting immediate and dynamic comprehension.
- ✓ **Scalability and Adaptability:** Design a flexible system capable of accommodating diverse classroom scenarios and adapting to various subjects and linguistic variations.
- ✓ **Efficiency and Responsiveness:** Ensure streamlined processing and minimal latency during the transformation stages, delivering a responsive user experience.
- ✓ **Integration with Educational Resources:** Enable seamless integration with educational materials, facilitating the transformation of class lessons, lectures, and educational content into sign language animations effortlessly.

3.4. Deployment Diagram

The deployment diagram featured in our project documentation offers a concise depiction of how the software components are distributed across hardware nodes or devices. It showcases the physical deployment of system elements, illustrating the allocation of components to specific hardware entities, such as servers, devices, or nodes. This diagram serves as a visual guide to understand the system's deployment architecture, portraying the relationship between software elements and the physical infrastructure where they operate.

❖ **Microphone:**

The microphone component represents the input source for capturing spoken language. It serves as the initial point of interaction, allowing users to verbally communicate with the system. The microphone captures audio data, which is then processed for further conversion.

❖ **User Interface:**

The user interface component is the gateway for user interaction with the system. It includes elements such as buttons, controls, and feedback mechanisms that facilitate a seamless user experience. Users can initiate voice input, control settings, and receive visual feedback through the user interface.

❖ **Voice to Text:**

The voice-to-text component is responsible for converting spoken words from the microphone into written text. This process involves speech recognition algorithms that analyze and transcribe the audio input into a textual format. The resulting text serves as the intermediary data for the subsequent stages of the system.

❖ **Text to Sign:**

The text-to-sign component translates the textual representation of spoken language into sign language. It employs natural language processing (NLP) and sign language generation algorithms to interpret the meaning of the text and convert it into corresponding sign language gestures or animations. This component ensures accurate and culturally sensitive sign language representation.

❖ **Display Screen:**

The display screen component is responsible for presenting the sign language animations generated from the text. It showcases the educational content in real-time, making it accessible to deaf students. The display screen may include a graphical user interface (GUI) that enhances the user experience with additional features such as playback controls and visualization options.

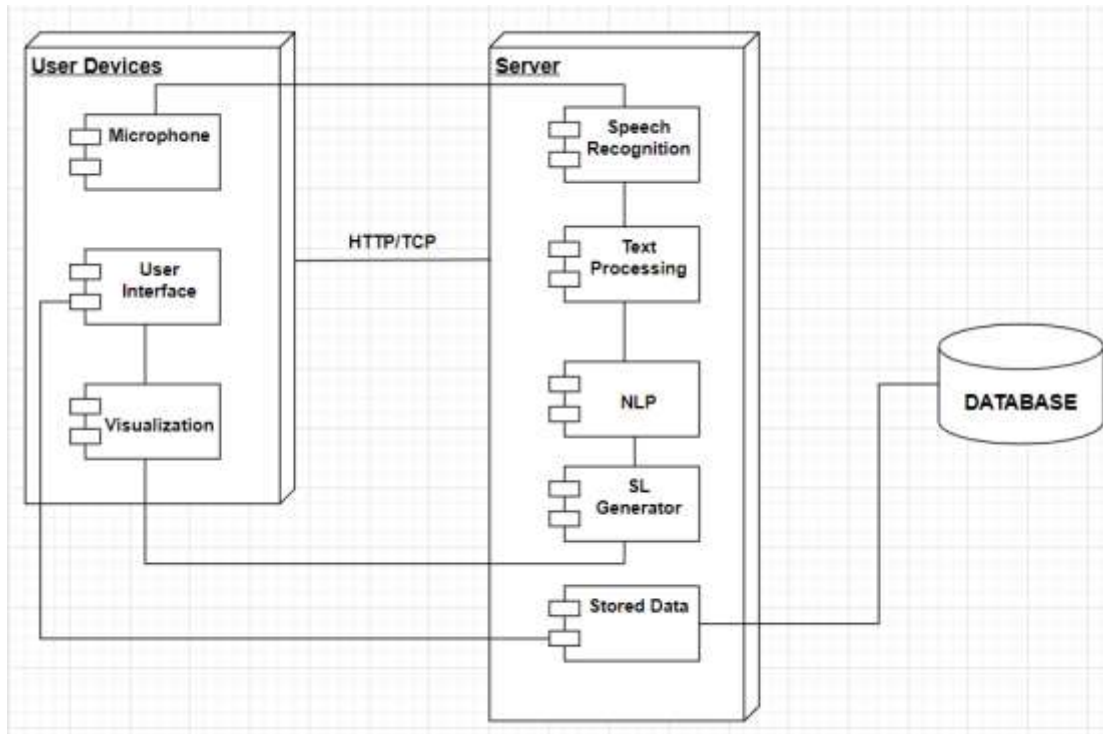


Figure 8 deployment diagram

3.5. Architectural Design

In the architectural design phase of this innovative project, the focus revolves around structuring a robust and cohesive framework that orchestrates the seamless transformation of spoken language into text and dynamic sign language animations. This phase encompasses the blueprint for the system's structure, outlining the key components, interactions, and technological infrastructure necessary to bring the envisioned platform to life.

3.5.1. Subsystem decomposition

Subsystem decomposition refers to the process of breaking down a complex system into smaller, more manageable and interconnected subsystems or modules. Each subsystem encapsulates specific functionalities or tasks within the larger system. This decomposition allows for a clearer understanding of the system's architecture, enhances manageability, promotes scalability, and facilitates efficient development and maintenance. It involves identifying the major components of the system and organizing them hierarchically based on their inter-dependencies and functionalities.

➤ Decomposed Subsystems:

Speech Input Subsystem:

- ✓ *Microphone Interface*: Manages the interaction with the microphone, capturing audio input from users.
- ✓ *Audio Processing Module*: Responsible for preprocessing and cleaning the raw audio data before passing it to the voice-to-text subsystem.

User Interaction Subsystem:

- ✓ *User Interface Module*: Handles the graphical user interface, providing controls for initiating voice input, adjusting settings, and displaying feedback to users.

Speech-to-Text Subsystem:

- ✓ *Speech Recognition Engine*: Implements algorithms for converting spoken language into text, ensuring accurate transcription of the audio input.
- ✓ *Text Processing Module*: Cleans and processes the transcribed text for better accuracy and prepares it for translation.

Text-to-Sign Subsystem:

- ✓ *Natural Language Processing (NLP) Module*: Utilizes NLP techniques to understand the meaning of the transcribed text.
- ✓ **Sign Language Generation Engine**: Generates sign language animations or gestures based on the interpreted meaning of the text.

Display Subsystem:

- ✓ **Visualization Module**: Manages the display of sign language animations on the screen in real-time.
- ✓ **User Feedback Module**: Provides visual feedback to users, such as progress indicators and completion messages.

3.5.2. Component diagram

The component diagram, a crucial visualization within our project documentation, succinctly portrays the structural composition of our system. It illustrates the high-level arrangement of various software components or modules and their interconnections, offering a clear depiction of the system's architecture. This diagram provides a concise and insightful overview, showcasing how individual components collaborate and interact to deliver the system's functionalities and features.

Key Components:

- ✓ **Microphone Input Component:** Represents the functionality to capture voice input from the microphone.
- ✓ **User Interface Component:** Provides the interface for user interaction and seamless integration of functionalities.
- ✓ **Speech-to-Text Component:** Converts the voice input to text.
- ✓ **Text-to-Sign-Language Component:** Transforms the text into sign language animations.
- ✓ **Display Component:** Shows the sign language animations on the screen in real-time.
- ✓ **Database Component:** Stores data crucial for the system's operation and access like: User accounts.

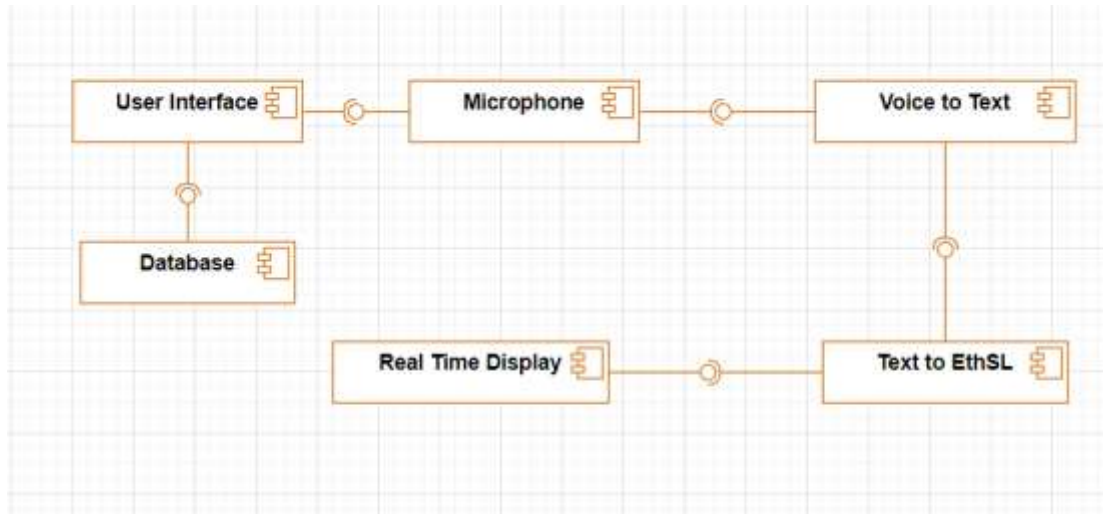


Figure 9 component diagram

3.5.3. Persistent modeling

Persistent modeling involves designing the structure for how data will be stored persistently in the system. The persistent modeling might include:

- ✓ **Data Storage for User Information:** Structuring how user profiles, preferences, and authentication data will be stored.
- ✓ **Machine Learning Models:** Storing the machine learning models or the data required for these models. This might involve defining how the model parameters or trained models will be saved.
- ✓ **Data Persistence Strategy:** Deciding how frequently data will be updated or stored, handling backups, and managing data integrity

3.5.4. Access Control and Security

Securing user data and ensuring the integrity of the system are essential considerations for this web application. This section delves into the robust access control and security measures implemented to build a fortress of trust for both hearing-impaired users and educational institutions.

Authentication and Authorization:

- ✓ **User Accounts:** All users, including students, lecturers, and administrators, require unique accounts with strong passwords or multi-factor authentication for secure access.
- ✓ **Role-Based Access Control (RBAC):** User roles define access privileges. Students can only access translated lectures, while lecturers can upload lecture recordings and manage settings. Administrators have full control over system configurations and user accounts.

Data Security:

- ✓ **Data Encryption:** Sensitive data, such as lecture recordings and translation logs and process, is encrypted at rest and in transit using secure algorithms.

Network Security:

Secure Sockets Layer (SSL/TLS): All communication between user devices and the web server is encrypted using HTTPS to protect against data interception and tampering.

Security Practices:

Principle of Least Privilege: Users are granted only the minimum access required to fulfill their tasks, minimizing the potential impact of compromised accounts.

Secure Coding Practices: Developers employ secure coding practices to avoid vulnerabilities in the system software.

Regular Security Audits and Penetration Testing: Independent security audits and penetration testing identify and address security weaknesses before they are exploited.

By implementing these comprehensive access control and security measures, the web application becomes a trusted platform for educational accessibility, safeguarding user data, and ensuring the integrity of translated sign language content.

3.5.5. Global Software Control

The architectural design incorporates a centralized control mechanism to efficiently manage and coordinate various processes within the system. The global software control aspect involves the following components:

Centralized Command Center:

- Centralized command center overseeing system processes and functionalities.
- Ensures uniformity in the execution of translation tasks and user interactions.

Control Mechanisms:

Event-Driven Architecture: Communication between modules occurs through synchronous events. Upon completion of a task, simultaneously a module publishes an event, notifying other modules that it requires specific data or functionalities.

Message Queues: Events are placed in a queue, ensuring asynchronous processing and preventing bottlenecks. Each module listens for relevant events and takes appropriate actions.

Real-time Performance Optimization: Techniques like caching frequently used data or prioritizing high-latency components can further improve response times and deliver seamless translation experiences.

Monitoring and Logging: Comprehensive monitoring systems track system health, performance metrics, and user activity. Logs provide valuable data for debugging, performance analysis, and security audits.

Benefits of Robust Control

- **Scalability:** The system can handle increased user load and diverse lecture environments by efficiently allocating resources and prioritizing tasks.
- **Fault Tolerance:** If a module encounters an error, other modules can continue functioning, minimizing disruptions and ensuring uninterrupted translation.
- **Maintainability and Debugging:** Event-driven architecture simplifies troubleshooting by providing clear traces of data flow and module interactions.
- **Flexibility and Adaptability:** New modules or functionalities can be easily integrated into the system without significant disruption to existing operations.

3.5.6. Boundary Conditions

This section dissects the crucial concept of boundary conditions, outlining the interfaces and protocols that connect the application to users, external systems, and the environment.

User Interface Boundaries:

- **Input Formats:** Define the supported formats for receiving audio input, such as streaming audio, pre-recorded files, or microphone recordings.
- **Output Formats:** Specify the formats for delivering translated sign language outputs, including animation rendering, or text representations.
- **Accessibility Considerations:** Ensure all user interfaces are accessible for users with needs, including screen readers, alternative input devices, and various display settings.

External System Boundaries:

- **APIs:** If the application integrates with other systems, such as lecture repositories or learning management platforms, define the APIs and communication protocols for exchanging data and functionality.
- **Error Handling:** Establish mechanisms for handling errors and exceptions arising from interactions with external systems, providing informative feedback.

Environmental Boundaries:

- **Network Performance:** Determine the minimum required network bandwidth and latency for seamless operation, taking into account real-world user environments and potential limitations.
- **Device Compatibility:** Specify the hardware and software requirements for user devices to ensure compatibility with the application's functionalities.

This ensures seamless exchange of information, robust functionality, and a secure and reliable user experience, ultimately contributing to the successful implementation of this valuable accessibility tool.

3.5.7. Machine learning model design

- ✓ **Data Collection:** Collect a large dataset of audio recordings of spoken words and phrases in the language you want to translate.
- ✓ **Preprocessing:** Preprocess the audio data by converting it into a format that can be used by machine learning models. This can include steps such as noise reduction, normalization, and feature extraction.
- ✓ **Speech-to-Text:** Train a speech-to-text model to convert the audio data into text. Training deep learning techniques such as convolutional neural networks (CNNs).
- ✓ **Text-to-Sign Language:** Train a sign language animation model to convert the text into sign language animations, by deep learning techniques such as CNNs.
- ✓ **Integration:** Integrate the speech-to-text and text-to-sign language models into a single pipeline that takes audio as input and produces sign language animations as output.
- ✓ **Testing and Evaluation:** Test the pipeline on a separate dataset to evaluate its performance.

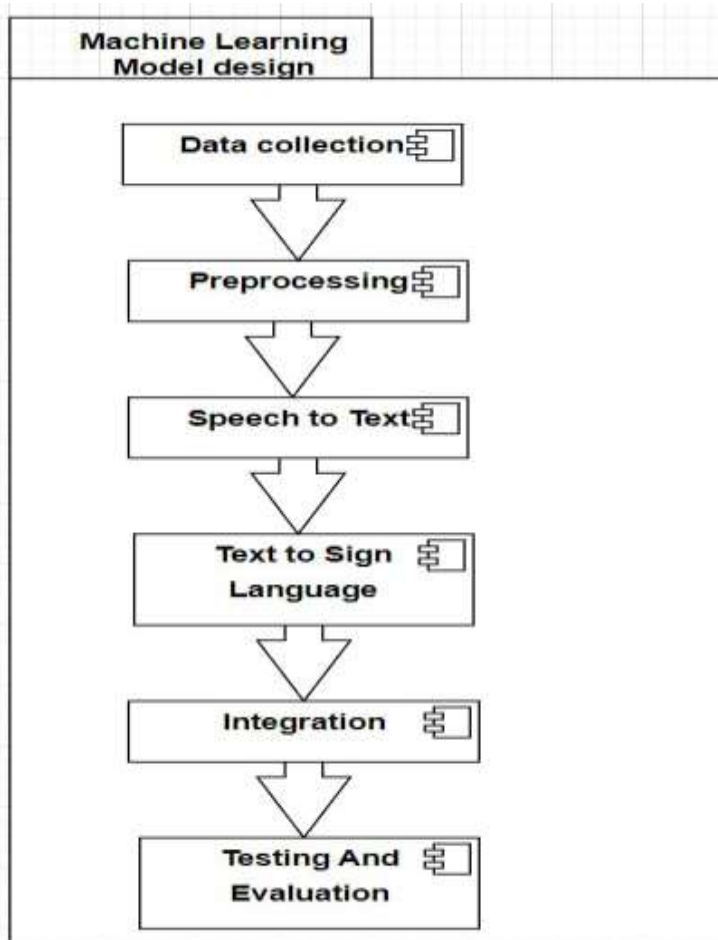


Figure 10 machine learning model design

3.6. User Interface Design

Logical Characteristics

The user interface design focuses on creating an intuitive and accessible experience for hearing-impaired individuals interacting with the web application. Key aspects include:

- **User-Friendly Navigation:** Intuitive navigation to ensure ease of use, with clear menus and options.
- **Responsive Design:** Ensuring the web application adapts seamlessly to various screen sizes and devices.

Sample Screen

Providing a visual representation of the anticipated user interface , including:

- **Homepage:** Featuring accessible options for speech-to-text and sign language translation.
- **Log in page:** Allow users to login or create account.
- **User Profile Page:** Allowing users to customize profile.
- **Feedback Page:** used to take a feedback from user in text form.

GUI Standards

Adherence to established GUI standards for a consistent and familiar user experience, encompassing:

- **Button Placement:** Consistent placement of buttons for actions like starting/stopping translation.
- **Feedback Mechanisms:** Clear text feedback to confirm user actions and system responses.
- **Standardized Icons:** Use of universally recognized icons for common functions, enhancing user understanding.

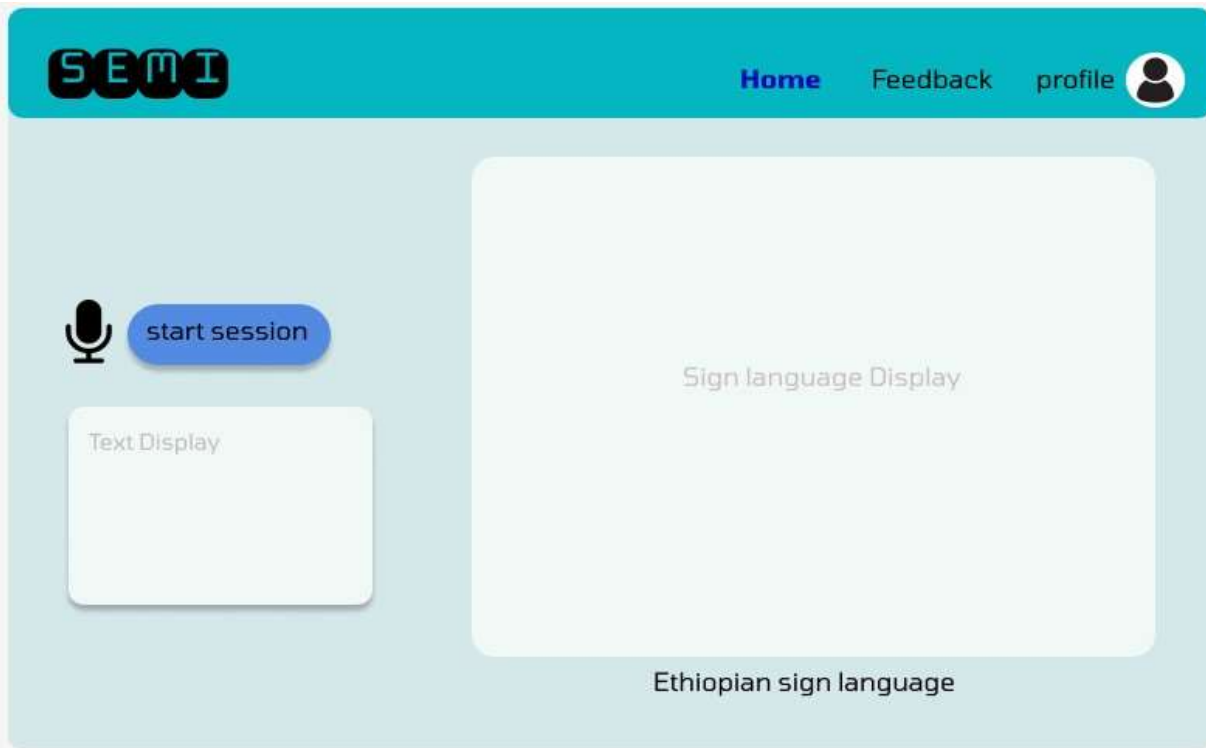


Figure 11 home page default display

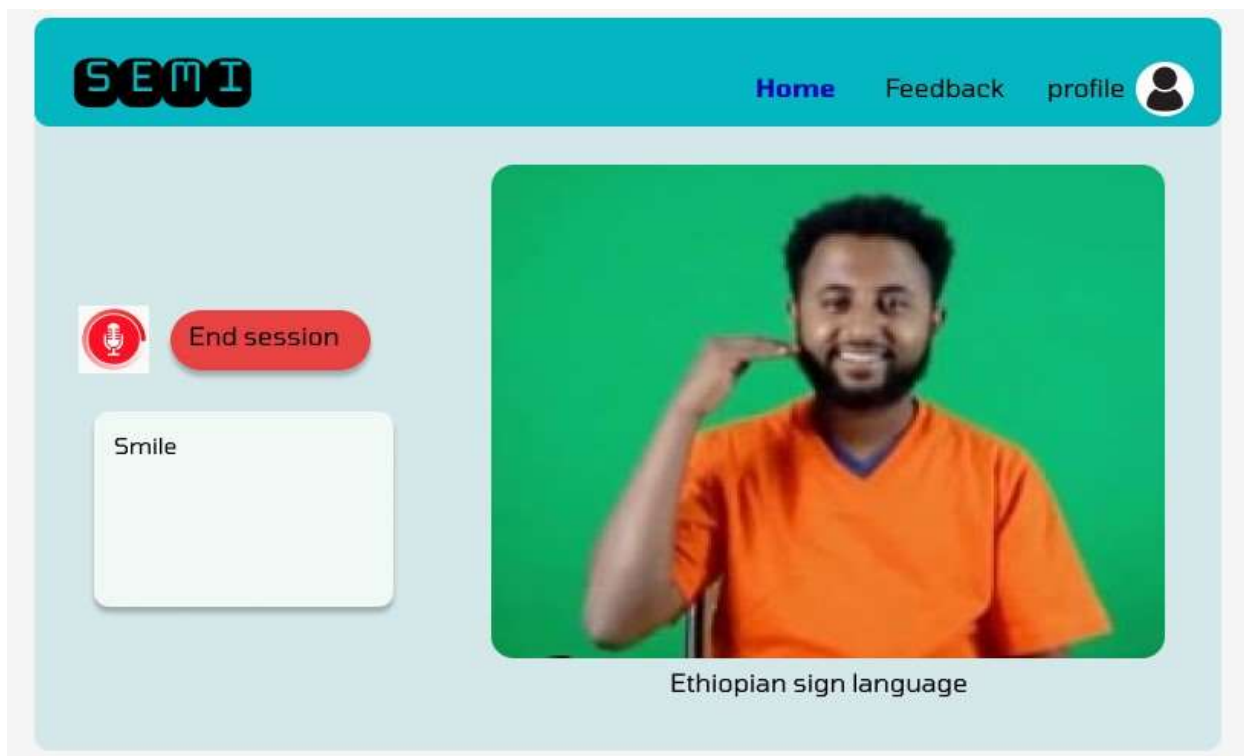


Figure 12 home page display when recording

The screenshot shows the SEMI Feedback page. At the top is a teal header with the SEMI logo on the left and navigation links for Home, Feedback, and Profile on the right. The main content area has a light blue background. On the left, there is a user profile section for Samuel Teshome, featuring a circular profile icon and the text 'Rate the platform helpfulness'. Below this are four buttons: 'Very helpful' (green), 'Good' (blue), 'Not Bad' (yellow), and 'Bad' (red). At the bottom of this section is a green 'send feedback' button. To the right of the profile section is a large, light gray rectangular box with the placeholder text 'write your feedback here'.

Figure 13 feedback page

The screenshot shows the SEMI Login page. It has a light gray background with a central white login form. The form is titled 'Login' at the top. It contains two input fields: 'Email' with the placeholder 'example@email.com' and 'Password' with a masked password '*****'. To the right of the password field is a blue link for 'Forgot password?'. Below the input fields is a prominent blue button labeled 'Login'. At the bottom of the form, there is a link that says 'Don't have account? Create new account'.

Figure 14 Login page

3.7. Database Design

In the database design since Python uses an object-relational mapping (ORM) system to interact with databases so in that case there is no need for ER diagram. The ORM system in Django allows developers to interact with databases using Python code and abstracts away many of the complexities of database management.

Furthermore, we can see the design in detail and the data structures to be used in the implementation and the table structure of all databases including full field descriptions in class diagram.

3.8. Machine learning algorithm

Machine learning algorithms play a crucial role in our project, enabling the transformation of voice inputs into sign language. To achieve voice recognition, we utilize an algorithm that accurately converts Amharic language into text. Additionally, we develop an algorithm that translates the text into sign language. These algorithms leverage neural networks to understand the intricacies of sign language and generate meaningful signs that correspond to the input text.

The integration of these algorithms is essential for ensuring real-time functionality and seamless operation. We continuously enhance the algorithms using machine learning techniques, enabling them to adapt to various languages and user interactions. This iterative approach allows us to create a system that delivers accurate and contextually relevant translations of voice inputs into sign language.

3.9. Summary

Final Year Project 1 focused on the conceptual development of a voice-to-sign language translation system, “ሰሚ” | “SEMI”, to address the communication challenges faced by the hearing-impaired community in Ethiopia. The project aimed to propose a theoretical framework for bridging the gap between spoken language and sign language using machine learning algorithms.

The project involved extensive research on voice recognition and text-to-sign language translation algorithms, with a focus on understanding their potential applications in the context of the hearing-impaired community. The theoretical framework of the “ሰሚ” | “SEMI” system was carefully designed, outlining the necessary components and algorithms required for voice-to-sign language translation.

Future Work

In the next phases of the project, the focus will shift towards the practical implementation of the proposed framework. Key areas of future work include:

Algorithm Development: - The voice recognition and text-to-sign language translation algorithms will be further explored and refined to improve accuracy and contextual understanding.

Prototype Development: - A software prototype will be developed to demonstrate the functionality and feasibility of the “ሰሚ” | “SEMI” system, allowing for testing and evaluation.

User Testing and Feedback: - Extensive user testing will be conducted with the hearing-impaired community to gather feedback and insights, ensuring that the system meets their specific needs and preferences.

Through continued research, algorithm refinement, and user testing, this project aims to deliver a practical and efficient solution that promotes inclusivity and empowers the hearing-impaired community to engage in meaningful conversations. With further advancements and collaboration, “ሰሚ” | “SEMI” has the potential to transform the lives of the hearing impaired, enabling them to participate more fully in various aspects of society.

References

- [Deafness and hearing loss \(who.int\)](#)
- <https://www.yumpu.com>
- Anderson, L., & Davis, M. (2019). "Effective Communication Strategies for the Hearing-Impaired in Educational Settings." *Journal of Educational Technology*, 22(4), 367-382.
- <https://www.researchgate.net>
- <https://www.academia.edu>
- <https://independent.academia.edu/LilyAbebe>
- <https://ftkki.umt.edu>
- <https://www.yumpu.com>
- National Association for the Deaf. (2020). "Educational Inclusivity and the Rights of Hearing-Impaired Students."
 - Retrieved from [[Ethiopian National Association of the Deaf \(ENAD\)](#) | [Ethiopian National Disability Action Network \(ENDAN\) \(endanethiopia.org\)](#)].
- World Health Organization. (2018). "Deafness and Hearing Loss."
 - Retrieved from [[Deafness and hearing loss \(who.int\)](#)].